

Amira Benslama

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EDUCATION

University of Cambridge | *MSc Machine Learning and Machine Intelligence* 2019–2021

- Thesis on quantized neural networks for edge inference achieved 94.2% accuracy with 8.7x model compression on ImageNet; published at ICLR 2021 Workshop
- President of Cambridge AI Society (250+ members); organized 12 seminars with speakers from DeepMind, Google Brain; coordinated £45K sponsorship funding

ETH Zurich | *BSc Computer Science with Systems Focus* 2015–2019

- GPA 3.89/4.0 with distinction in all machine learning courses; awarded ETH Medal for exceptional thesis on distributed training systems
- Led systems lab teaching assistant role for 120+ students; mentored 8 junior researchers on CUDA optimization, improving their codebases by avg. 3.2x throughput

EXPERIENCE

Stripe | *Senior Machine Learning Engineer, Inference Pipelines* 2022–Present

- Architected real-time fraud detection inference pipeline processing 12.8M transactions/day; reduced latency from 580ms to 94ms (6.2x improvement) through model distillation and ONNX optimization, saving \$2.3M/year in compute costs
- Led cross-functional team of 5 ML engineers and 3 infrastructure engineers; designed dynamic batching system that increased GPU utilization from 38% to 81% while maintaining P99 latency under 120ms
- Open-sourced TorchServe feature for quantized INT8 inference that achieved 4.1M downloads; contributed to 2.8 GitHub stars from community adoption across 180+ companies

Datadog | *Machine Learning Engineer, ML Platform* 2021–2022

- Optimized anomaly detection model serving infrastructure; migrated from TensorFlow 1.x to PyTorch + TorchScript, achieving 8.3x throughput improvement (2.1M inferences/sec to 17.4M inferences/sec) on CPU-only clusters
- Implemented automated A/B testing framework for inference model updates; reduced deployment risk and enabled 3x faster iteration cycles; measured 23% uplift in false positive reduction across 50+ customer accounts
- Mentored 4 junior ML engineers on production systems; established inference performance optimization best practices adopted org-wide, reducing median model latency by 31% across 140+ production models

Palantir Technologies | *Machine Learning Engineer, Inference Systems* 2019–2021

- Built distributed inference orchestration layer for graph neural networks on enterprise datasets; scaled throughput from 42K to 340K inferences/sec while maintaining <180ms P99 latency across 8-GPU clusters
- Developed model serving gateway with intelligent request routing that reduced tail latency (P99) by 47% through dynamic load balancing and speculative execution strategies
- Contributed to internal ML model registry and version control system; implemented automated performance regression testing that caught model degradations before production, reducing incidents by 56% YoY

Jane Street | *Quantitative Researcher, ML Infrastructure* 2018–2019

- Designed ultra-low latency inference system for reinforcement learning trading models; achieved <2.3ms end-to-end inference latency through custom C++ runtimes and SIMD optimizations on x86 hardware
- Optimized neural network model compression pipeline reducing model size from 890MB to 12MB (74x reduction) via knowledge distillation and pruning while maintaining 99.4% prediction accuracy

- Established inference performance benchmarking framework tracking 18 production models across GPU/CPU/FPGA targets; provided weekly reports guiding resource allocation decisions saving ~\$1.8M quarterly in hardware costs

PROJECTS

QuantumSpark | *Python, ONNX, TensorRT, Kubernetes, Redis* 2021–Present

- Open-source framework for deploying quantized deep learning models at scale; 8.2K GitHub stars and 142K monthly PyPI downloads; benchmarks show 5.8x throughput improvement over standard ONNX Runtime
- Implemented adaptive quantization algorithm that automatically determines optimal precision per layer; reduces model size by avg. 6.3x with <0.8% accuracy loss validated on 47 model architectures

PipelineOptima | *Go, Protocol Buffers, gRPC, PostgreSQL, Prometheus* 2020–Present

- Distributed inference orchestrator with automatic request batching and model ensemble management; deployed in production serving 890K requests/sec at Datadog and Stripe partners; 3.1K GitHub stars
- Reduces inference latency variance by 64% through priority queue scheduling and dynamic request timeout adjustment; enables 99.8% SLA compliance on variable workloads

EdgeThreshold | *TensorFlow Lite, ONNX, Rust, gRPC, Redis* 2019–2020

- Mobile and edge inference optimization toolkit achieving 11x model compression and 7.2x latency reduction on ARM processors; 2.3K GitHub stars and 58K downloads on pub.dev (Rust community)
- Case study published demonstrating on-device multi-lingual NLP inference (40 languages) with 680KB model size achieving 94.1% accuracy; adopted by 12 commercial mobile apps across 2.8M+ users

ServeCache | *Python, C++, CUDA, RDMA, FastAPI* 2018–2019

- High-performance model inference cache layer exploiting temporal and spatial locality; 1.9K GitHub stars; reduces compute by 73% on repetitive inference queries with <8ms cache lookup latency
- Benchmark shows 12.4x throughput improvement on transformer model serving under realistic traffic patterns; publicly validated on BERT, T5, and LLAMA family models

EXTRACURRICULAR

NeurIPS Conference | *Invited Speaker & Workshop Organizer* 2021–Present

- Delivered keynote 'Inference at Scale: From Edge to Enterprise' at NeurIPS 2023 and 2024 (1,200+ attendees each); published accompanying technical whitepaper with 890 citations
- Organized annual 'ML Serving & Deployment' workshop attracting 450+ industry practitioners; managed 8-person organizing committee and secured \$180K sponsorship from major tech companies

Open Source Community | *Maintainer & Mentor* 2020–Present

- Lead maintainer of 4 high-impact inference libraries (QuantumSpark, PipelineOptima, EdgeThreshold, ServeCache) with combined 15K+ GitHub stars; manage 23 active contributors and review 140+ PRs monthly
- Established mentorship program pairing 6 junior ML engineers from underrepresented backgrounds with industry mentors; 100% program completion rate with 5 mentees securing roles at tier-1 tech companies (Google, Microsoft, Meta)

TECHNICAL SKILLS

Languages – Python, C++, Rust, Go, SQL, CUDA

Technologies – PyTorch, TensorFlow, ONNX, TensorRT, TorchServe, Kubernetes, gRPC, Protocol Buffers, Redis, PostgreSQL, RDMA

Tools – Docker, Prometheus, Grafana, Git, JIRA, Jupyter, DVC, Weights & Biases, Kubeflow, Ray